Introduction To MATLAB®

المقـــــدمة

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Abu Deif

Introduction to MATLAB

MATLAB (short for MATrix LABoratory) is a special-purpose computer program optimized to perform engineering and scientific calculations. It started life as a pro gram designed to perform matrix mathematics, but over the years it has grown into a flexible computing system capable of solving essentially any technical problem.

The MATLAB program implements the MATLAB programming language and provides a very extensive library of predefined functions to make technical programming tasks easier and more efficient.

Rules on Variables and File Names

A variable can be named according to the following rules:

- Must begin with a letter.
- Can be up to 63 characters long.
- Can contain letters, digits, and the underscore character.
- Cannot contain punctuation characters (e.g., period, comma, semicolon).
- MATLAB is case sensitive: it distinguishes between uppercase and lowercase letters. For example, AA, Aa, aA, and aa are the names of four different variables.
- No spaces are allowed between characters (use the underscore where a space is desired).
- Avoid using the name of a built-in function for a variable (i.e., avoid using cos, sin, exp, sqrt, etc.). Once a function name is used to define a variable, the function cannot be used.

Predefined Variables and Keywords

There are words, called keywords, that are reserved by MATLAB for various purposes and cannot be used as variable names. These words are:

break case end for catch classdef continue else function global persistent return spmd if switch elseif otherwise parfor try while pi

MATLAB®

واجهة البرنامج

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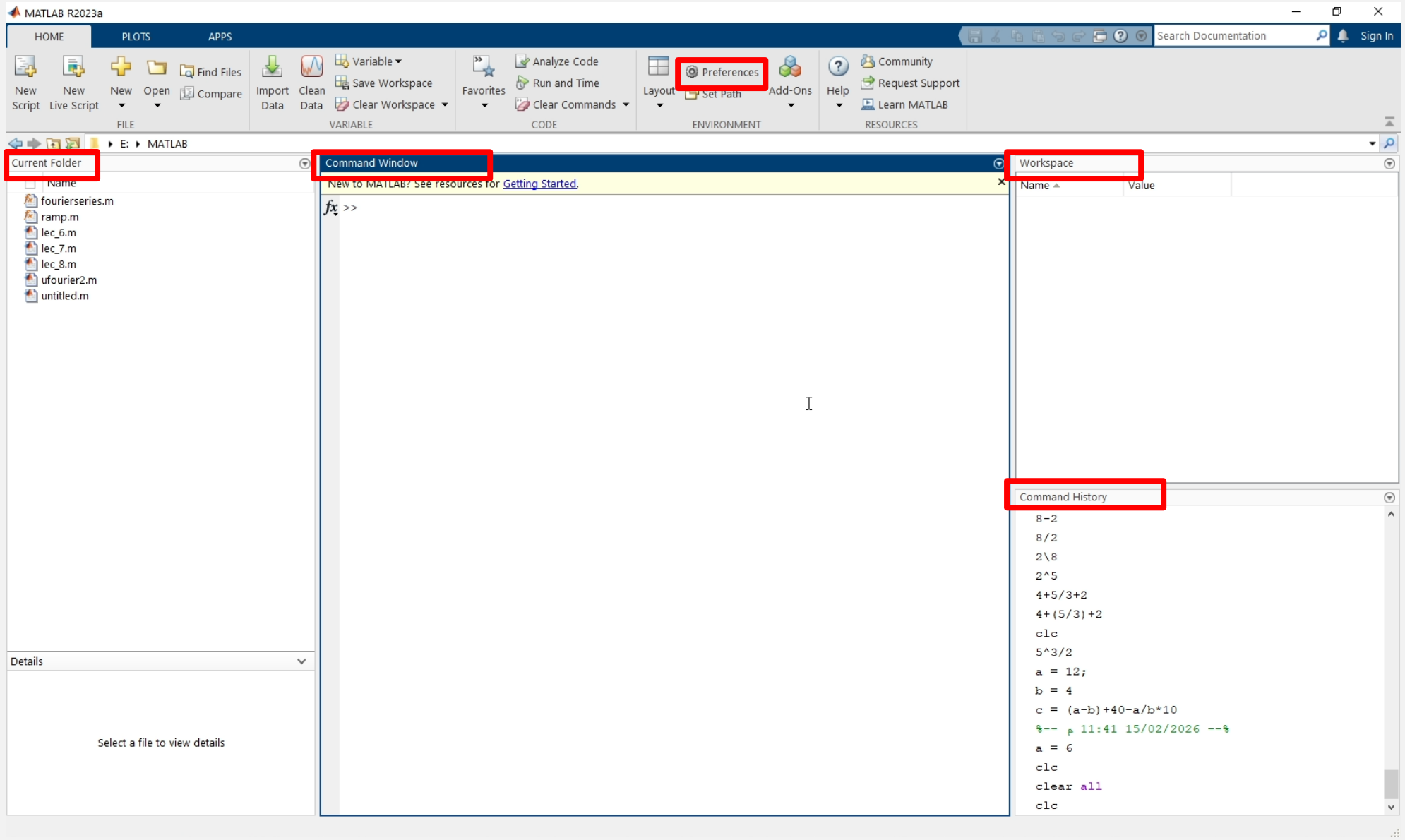
The MATLAB Environment



MATLAB®

العمليات الحسابية

ARITHMETIC OPERATIONS

Order of Precedence

DISPLAY FORMATS

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Chapter 1 Starting with MATLAB

- ARITHMETIC OPERATIONS
- Order of Precedence
- Using MATLAB as a Calculator
- DISPLAY FORMATS
- USEFUL COMMANDS FOR MANAGING VARIABLES
- ELEMENTARY MATH BUILT-IN FUNCTIONS
- EXAMPLES OF MATLAB APPLICATIONS
- PROBLEMS

ARITHMETIC OPERATIONS WITH SCALARS

Operation	Symbol	Example
Addition	+	5 + 3
Subtraction	–	5 – 3
Multiplication	*	5 * 3
Right division	/	5 / 3
Left division	\	5 \ 3 = 3 / 5
Exponentiation	^	5 ^ 3 (means $5^3 = 125$)

ARITHMETIC OPERATIONS WITH SCALARS

1.3.1 Order of Precedence

MATLAB executes the calculations according to the order of precedence displayed below. This order is the same as used in most calculators.

Precedence	Mathematical Operation
First	Parentheses. For nested parentheses, the innermost are executed first.
Second	Exponentiation.
Third	Multiplication, division (equal precedence).
Fourth	Addition and subtraction.

Using MATLAB as a Calculator

```
>> 7+8/2
ans =
    11
>> (7+8)/2
ans =
    7.5000
>> 4+5/3+2
ans =
    7.6667
>> 5^3/2
ans =
   62.5000
>> 27^(1/3)+32^0.2
ans =
     5
>> 27^1/3+32^0.2
ans =
    11
```

Type and press **Enter**.

8/2 is executed first.

Type and press **Enter**.

7+8 is executed first.

5/3 is executed first.

5^3 is executed first, /2 is executed next.

1/3 is executed first, 27^(1/3) and 32^0.2 are executed next, and + is executed last.

27^1 and 32^0.2 are executed first, /3 is executed next, and + is executed last.

```
>> a=12
a =
    12
>> B=4
B =
     4
>> C=(a-B)+40-a/B*10
C =
    18
```

Assign 12 to a.

Assign 4 to B.

Assign the value of the expression on the right-hand side to the variable C.

```
>> a=12;
>> B=4;
>> C=(a-B)+40-a/B*10;
>> C
C =
    18
```

The variables a, B, and C are defined but are not displayed since a semicolon is typed at the end of each statement.

The value of the variable C is displayed by typing the name of the variable.

DISPLAY FORMATS

Command	Description	Example

`format short`	Fixed-point with 4 decimal digits for: $0.001 \le number \le 1000$ Otherwise display format `short e`.	`>> 290/7`
		`ans =`
		`41.4286`

`format long`	Fixed-point with 15 decimal digits for: $0.001 \le number \le 100$ Otherwise display format `long e`.	`>> 290/7`
		`ans =`
		`41.428571428571431`

`format short e`	Scientific notation with 4 decimal digits.	`>> 290/7`
		`ans =`
		`4.1429e+001`

`format long e`	Scientific notation with 15 decimal digits.	`>> 290/7`
		`ans =`
		`4.142857142857143e+001`

ELEMENTARY MATH BUILT-IN FUNCTIONS

```
>> sqrt(64)
ans =
     8
>> sqrt(50+14*3)
ans =
    9.5917
>> sqrt(54+9*sqrt(100))
ans =
    12
>> (15+600/4)/sqrt(121)
ans =
    15
>>
```

Argument is a number.

Argument is an expression.

Argument includes a function.

Function is included in an expression.

ELEMENTARY MATH BUILT-IN FUNCTIONS

Table 1-3: Elementary math functions

Function	Description	Example

`sqrt(x)`	Square root.	`>> sqrt(81)`
		`ans =`
		`9`

`nthroot(x,n)`	Real *n*th root of a real number *x*. (If *x* is negative *n* must be an odd integer.)	`>> nthroot(80,5)`
		`ans =`
		`2.4022`

`exp(x)`	Exponential (e^x).	`>> exp(5)`
		`ans =`
		`148.4132`

`abs(x)`	Absolute value.	`>> abs(-24)`
		`ans =`
		`24`

`log(x)`	Natural logarithm.
	Base *e* logarithm (ln).
		`>> log(1000)`
		`ans =`
		`6.9078`

`log10(x)`	Base 10 logarithm.	`>> log10(1000)`
		`ans =`
		`3.0000`

ELEMENTARY MATH BUILT-IN FUNCTIONS

Function	Description	Example

`sin(x)`
`sind(x)`
	Sine of angle x (x in radians).
	Sine of angle x (x in degrees).
		`>> sin(pi/6)`
		`ans =`
		`0.5000`

`cos(x)`
`cosd(x)`
	Cosine of angle x (x in radians).
	Cosine of angle x (x in degrees).
		`>> cosd(30)`
		`ans =`
		`0.8660`

`tan(x)`
`tand(x)`
	Tangent of angle x (x in radians).
	Tangent of angle x (x in degrees).
		`>> tan(pi/6)`
		`ans =`
		`0.5774`

`cot(x)`
`cotd(x)`
	Cotangent of angle x (x in radians).
	Cotangent of angle x (x in degrees).
		`>> cotd(30)`
		`ans =`
		`1.7321`

USEFUL COMMANDS FOR MANAGING VARIABLES

Command	Outcome
`clear`	Removes all variables from the memory.
`clear x y z`	Removes only variables `x`, `y`, and `z` from the memory.
`who`	Displays a list of the variables currently in the memory.
`whos`	Displays a list of the variables currently in the memory and their sizes together with information about their bytes and class (see Section 4.1).